

Linear Algebra – The Backbone of AI/ML

Linear Algebra – The

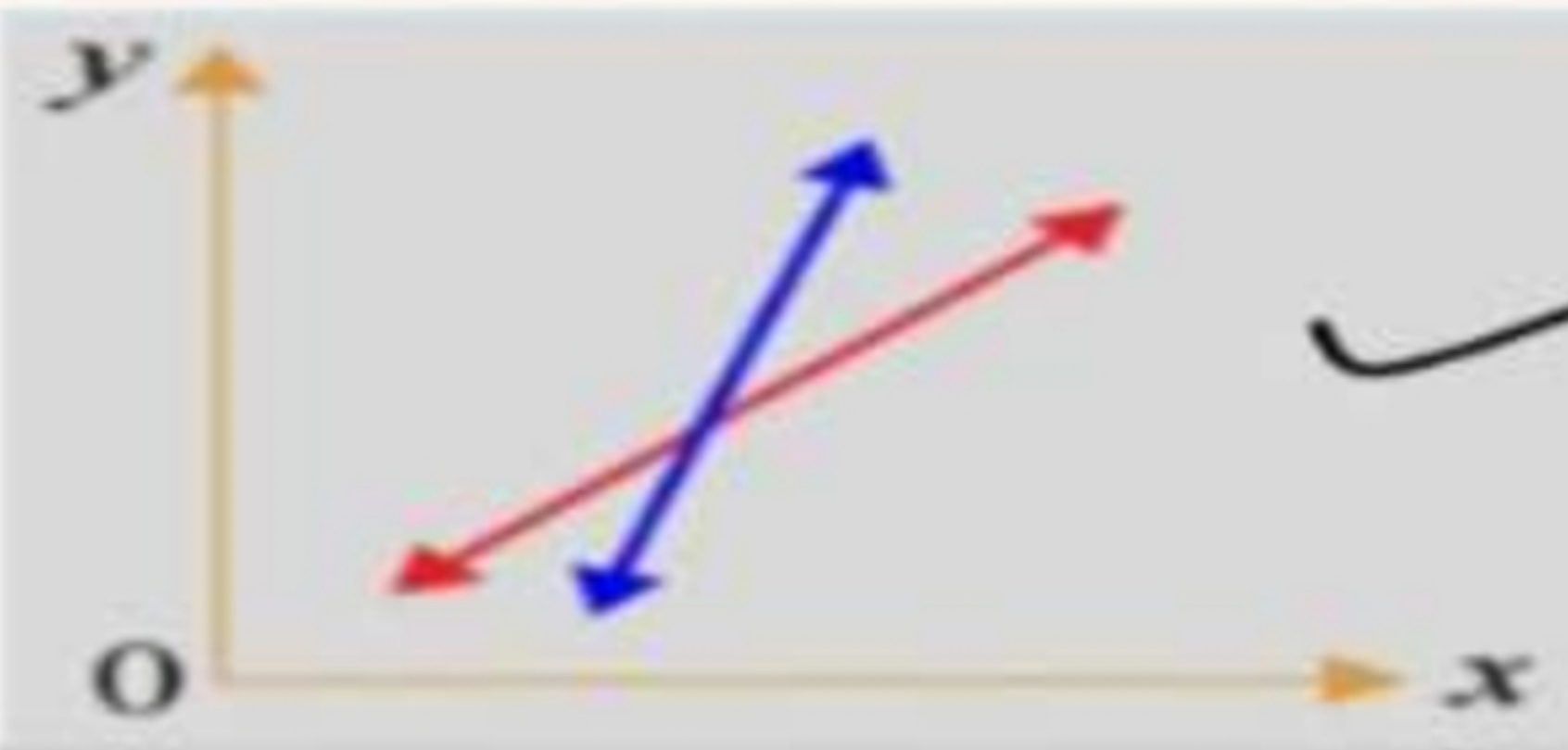
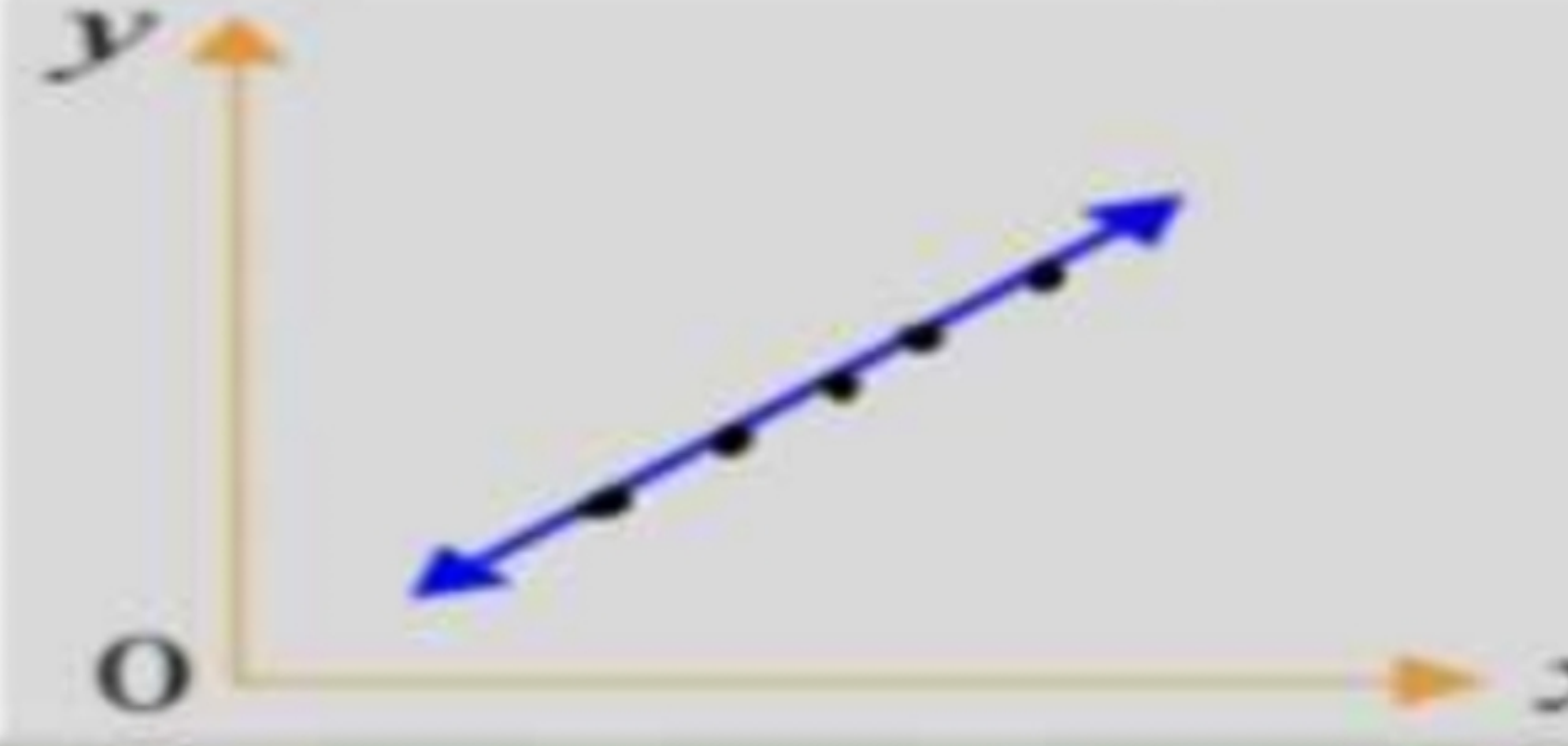
Sanjay Kumar Sen
Assistant Professor at
Quad AI



$$a_1x + b_1y + c_1 = 0$$

$$a_2x + b_2y + c_2 = 0$$

Nature of solution depending on the coefficient ratio :

Coefficient Ratio	Graphical Representation	Algebraic Interpretation
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	 Intersecting lines	Exactly one solution (unique)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	 Coincident lines	Infinitely many solutions
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	 Parallel lines	No solution



① The System of Linear Equation:

$$4x + 2y = 7$$

$$2x + y = 6$$

has -

(a) a unique solⁿ.

~~(b)~~ no solution

(c) an infinite no. of solⁿ

(d) exactly two distinct solⁿ -

$$a_1 = 4, b_1 = 2, c_1 = 7$$

$$a_2 = 2, b_2 = 1, c_2 = 6$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = 2 \quad \checkmark$$

$$\frac{c_1}{c_2} = \frac{7}{6}$$

$$\therefore \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$



Q.2

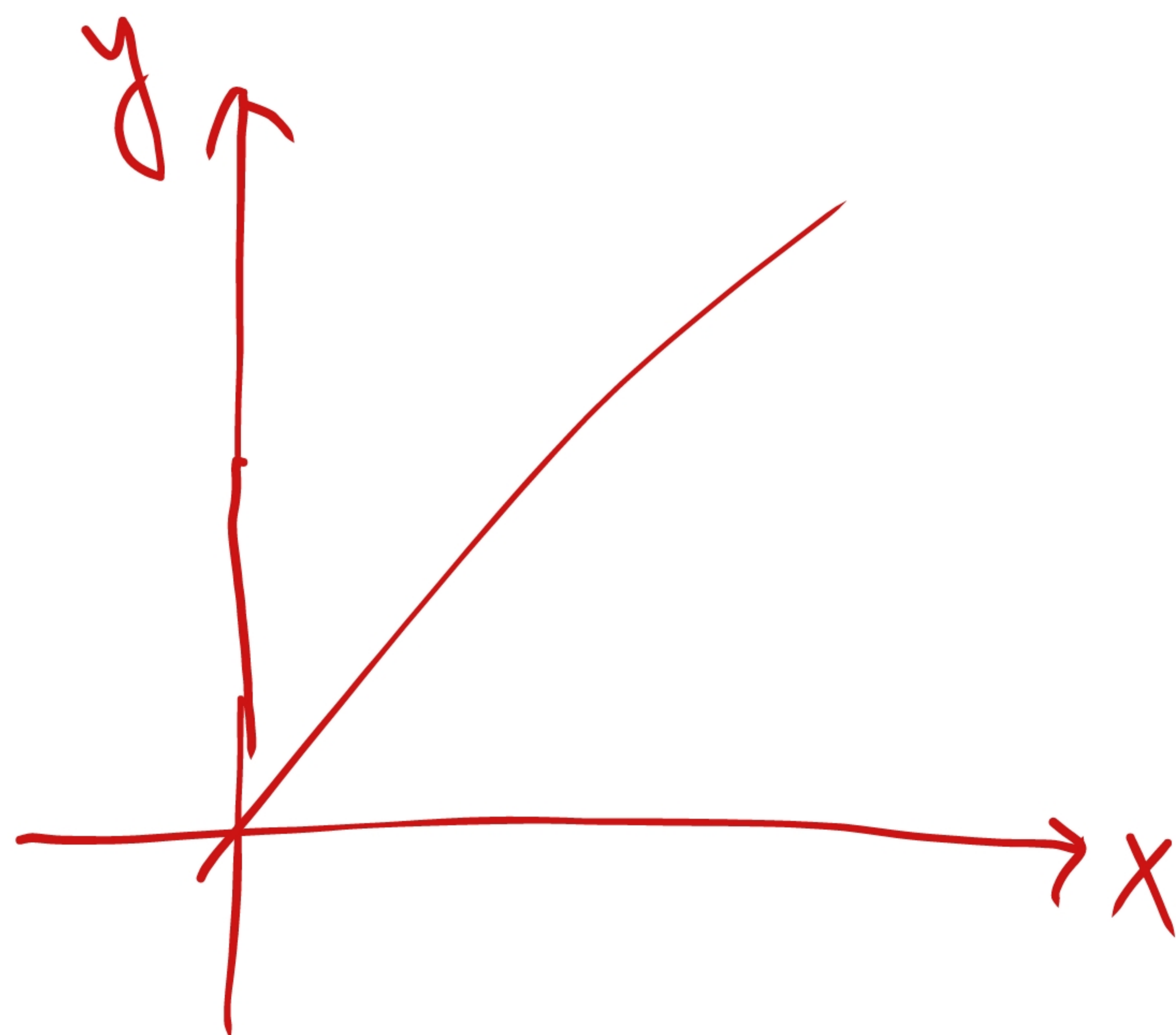
A straight line of the form $y = mx + c$ passes through the origin, and the point $(x, y) = (2, 6)$. Then the value of m —

$$c = 0 \text{ (given)}$$

$$y = mx$$

$$6 = 2m$$

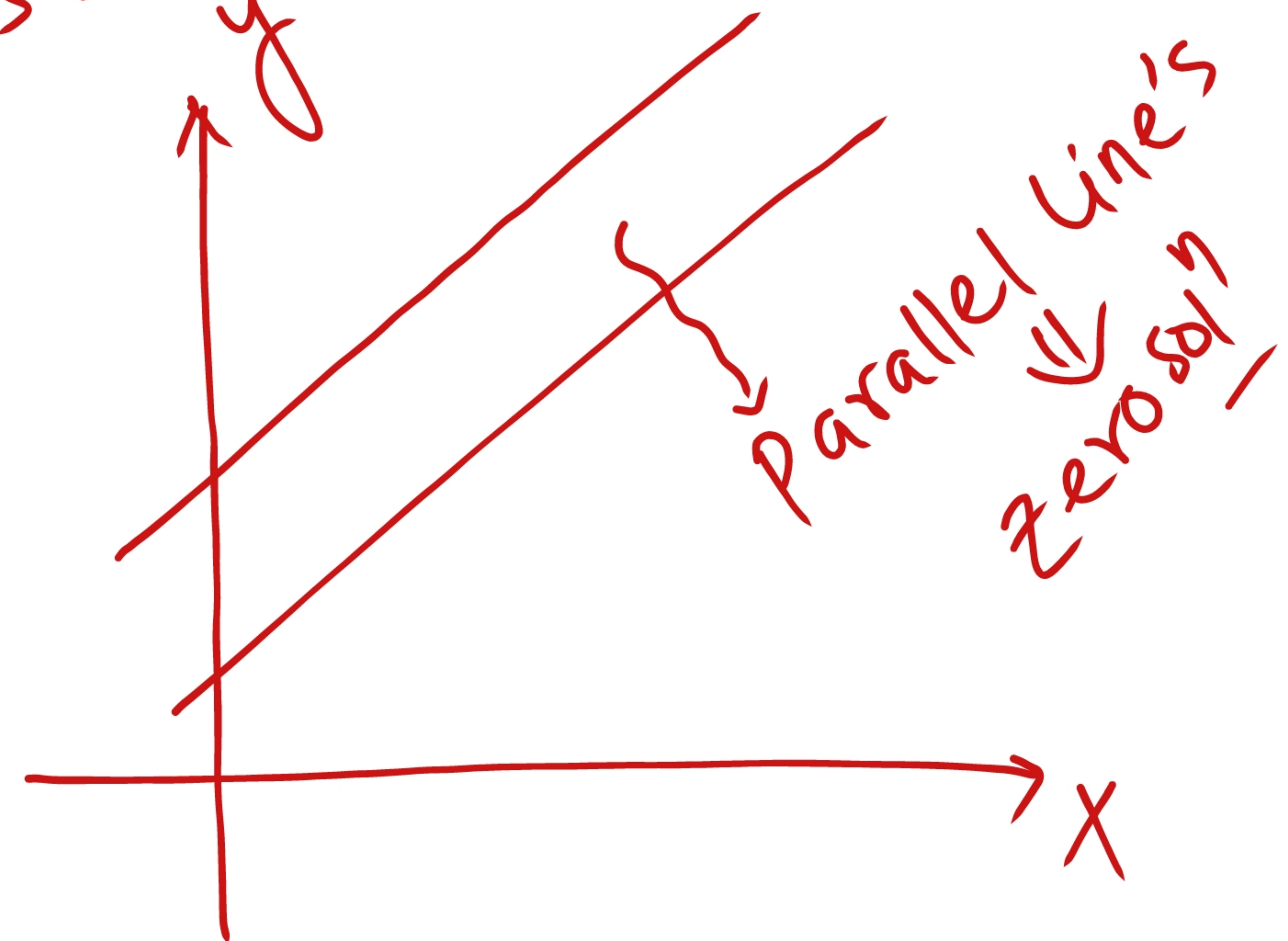
$$m = 3$$



Q.3 The no. of Solⁿs of the Simultaneous algebraic equations

$y = 3x + 3$ and $y = 3x + 5$ is -

- (A) zero
- (B) 1
- (C) 2
- (D) infinite



L.E of three var

$$\begin{aligned} x + y + z &= 24 \\ 5x + 3y + z &= 86 \\ x + y - z &= 0 \end{aligned}$$

$$\begin{aligned} x(x+y) &= 24 \\ x + y &= 12 \end{aligned}$$

$$\begin{aligned} 5x + 3y + z &= 86 \\ x + y + z &= 24 \\ \hline 4x + 2y &= 62 \\ 2x + y &= 31 \end{aligned}$$

$$\begin{aligned} 2x + y &= 16 \\ x + y &= 12 \\ \hline x &= 4 \\ y &= 8 \\ z &= 12 \end{aligned}$$

general eqn

$$\begin{cases} a_1x + b_1y + c_1 = d_1 \\ a_2x + b_2y + c_2 = d_2 \\ a_3x + b_3y + c_3 = d_3 \end{cases}$$



Linear Algebra

Conclusion :

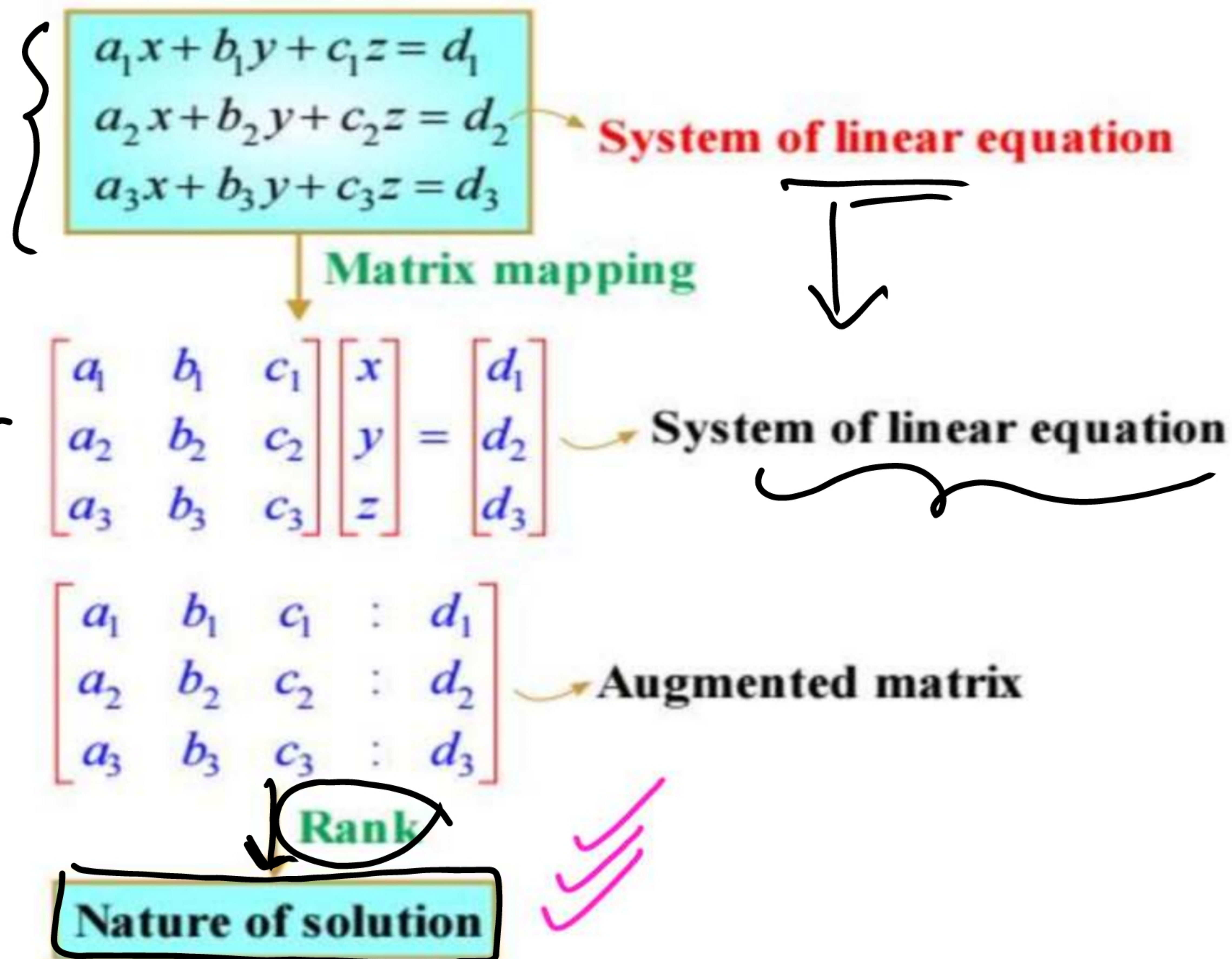
- (i) As number of variables, and number of equations increases. The solution becomes tedious and difficult to calculate. ✓
- (ii) Hence, the system of linear equations are mapped into matrix form and the rank of augmented matrix is calculated to decide the nature of solution. //

Linear Equation
 $2x + y = 10$
 $x + 3y = 20$

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \end{bmatrix}$$

$$\frac{2}{2} = \frac{4}{4}$$

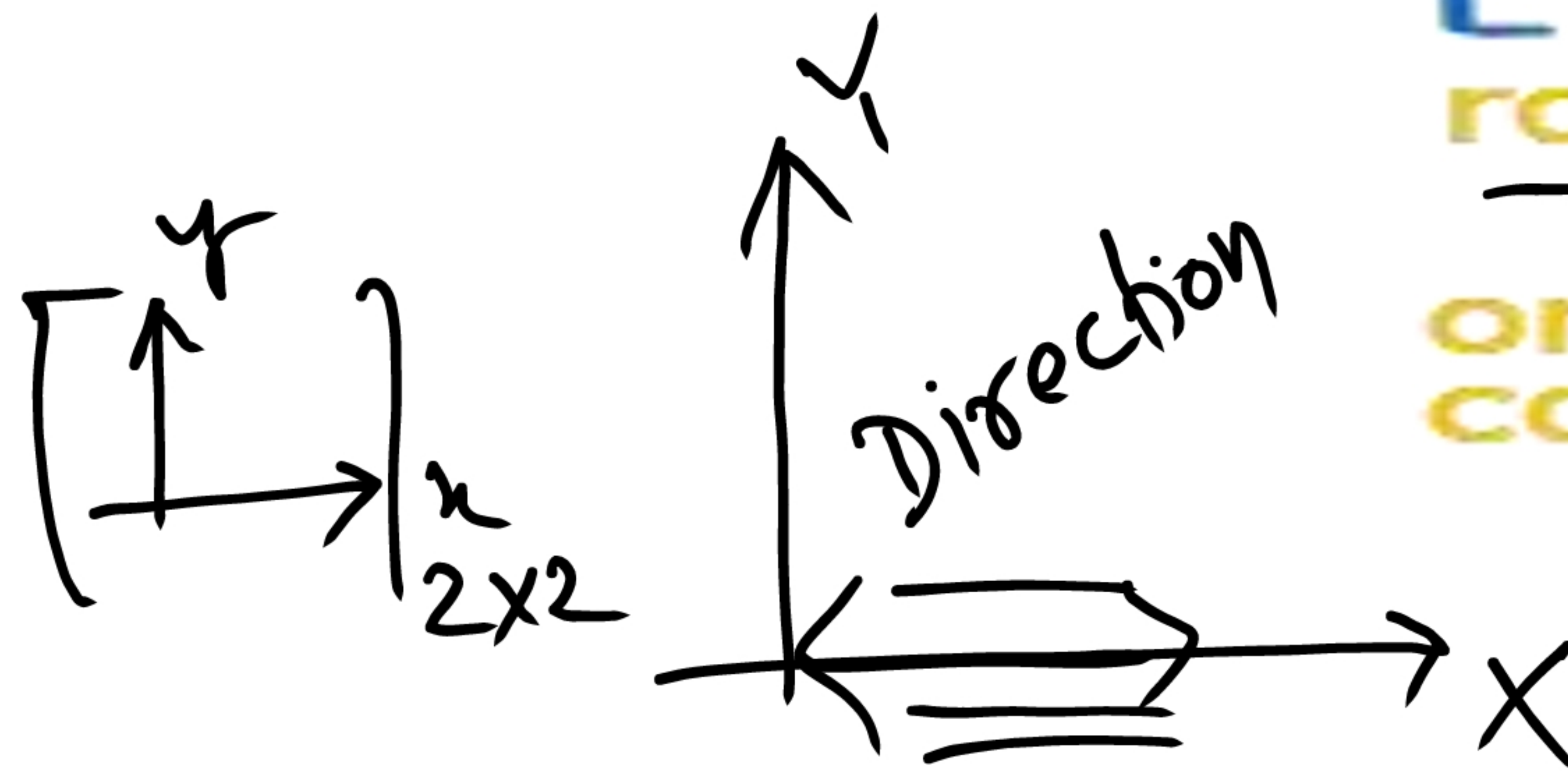
Matrix Mapping of System of Linear Equations



Vectors and Matrices

Scalar

24



Vector

$[2 \quad -8 \quad 7]$

row

or
column

$\begin{bmatrix} 2 \\ -8 \\ 7 \end{bmatrix}$

Matrix

$\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}$

row(s) × column(s)

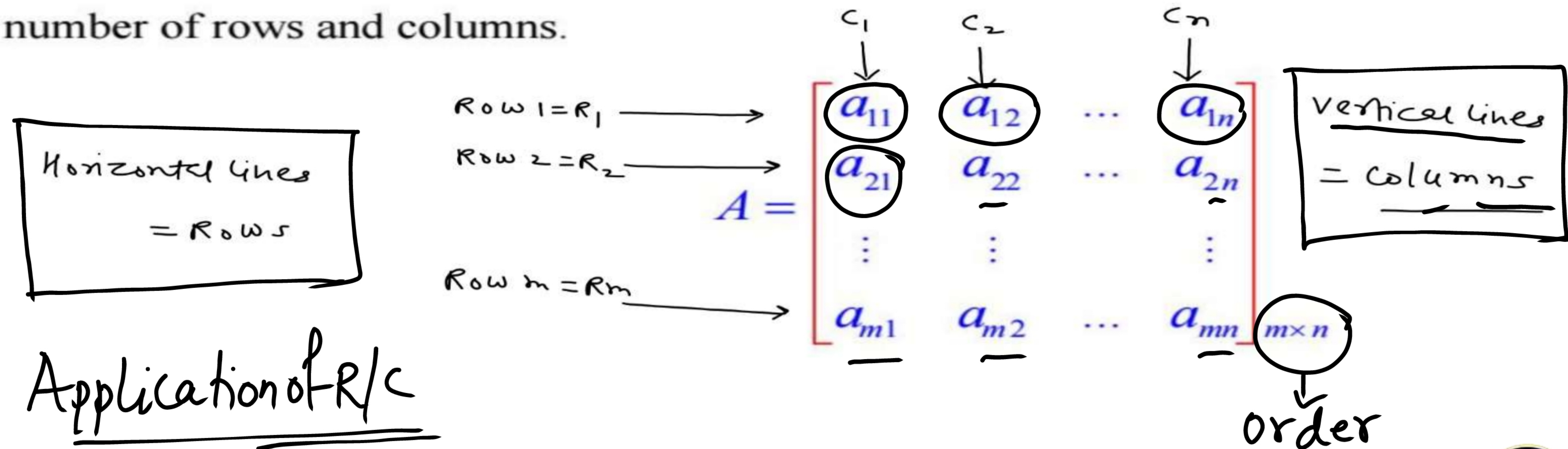
* combination of R/C.

- **Vector**:-If a matrix has only one row or only one column it is called a vector.
- **Matrix**:-A matrix is a two-dimensional array that has a fixed number of rows and columns and contains a number at the intersection of each row and column



Matrix and its Elements :

Matrix is the collection of numbers (Real, Imaginary or Complex) in fixed number of rows and columns.



Linear Algebra



Uses of LA

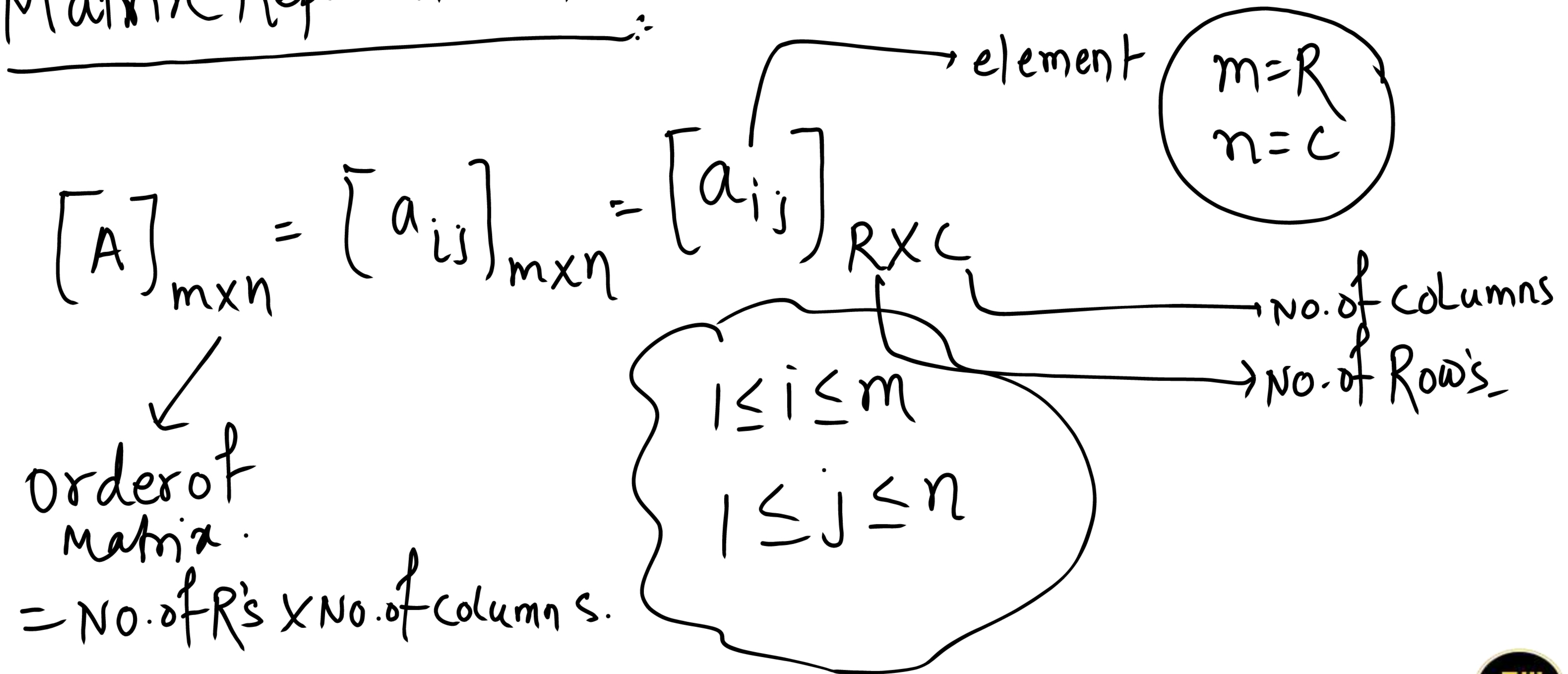


Linear Algebra



Application of column

Matrix Representation



$$[a_{ij}]_{2 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{matrix} \rightarrow R1 \\ \rightarrow R2 \end{matrix}$$

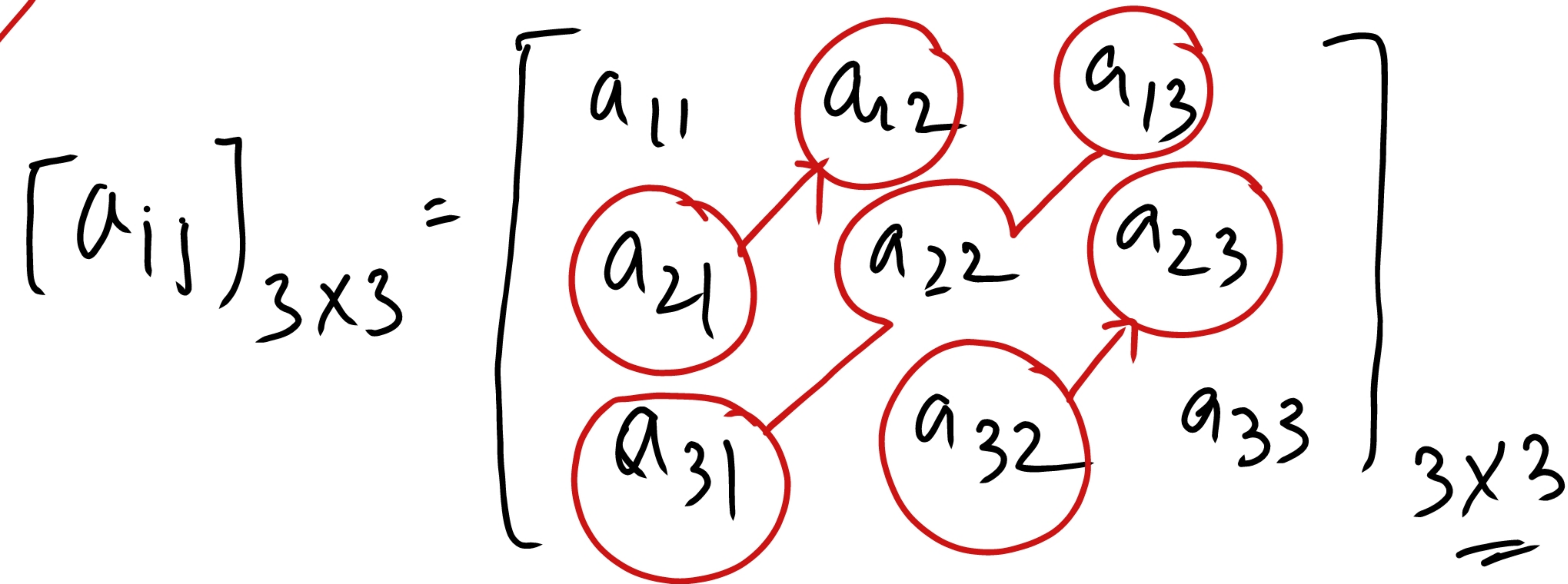
$$m = 2$$

$$n = 2$$

$$\begin{aligned} \text{No. of Elements} &= m \times n \\ &= n \times n \\ &= n^2 = 2 \times 2 = 4 \checkmark \end{aligned}$$



Conjugate :-



Conjugate
 $1 \rightarrow 2$
 $2 \rightarrow 1$
 $1 \rightarrow 3$
 $3 \rightarrow 1$

Note :- a_{ij} and a_{ji} are conjugate elements ✓
 $[a_{ij}]_{4 \times 4} = [A]_{4 \times 4}$
 No. of Elements = $n \times n = 4 \times 4 = 16$



Linear Algebra

Types of Matrix According to its Dimensions : ✓

1. Square Matrix : ✓

A matrix in which the number of rows are equal to the number of columns is known as a square matrix ($R = C$ or $m = n$).

✓ $A = [a_{ij}]_{m \times m} ; m = n$

For Example : $[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$

$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$

$$[a_{ij}]_{n \times n} = [A]_{n \times n}$$

No of elements = $m \times n$
 $= n \times n$
 $= n^2$

$= 2 \times 2$
 $= 4$



Linear Algebra

2. Rectangular Matrix :

A matrix in which the number of rows is not equal to the number of columns is known as a rectangular matrix ($R \neq C$ or $m \neq n$).

ex's

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3}$$

$$R \neq C \quad m \neq n$$

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \end{bmatrix}_{3 \times 4}$$



3. Row Matrix :

A matrix in which only one row and any number of columns is known as row matrix (or) row vector.

$$A = [1 \ 2 \ 3 \ 4] \checkmark \checkmark$$



4. Column Matrix :

A matrix in which only one column and any number of rows is called a column matrix (or) column vector.

$$B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Vaishu's
(Tech fest)
1) website
2) coding con,




```

# importing library sympy
from sympy import symbols, Eq, solve

# defining symbols used in equations
# or unknown variables
x, y = symbols('x,y')

# defining equations
eq1 = Eq((x+y), 10)
print("Equation 1:")
print(eq1)
eq2 = Eq((x-y), 20)
print("Equation 2")
print(eq2)

# solving the equation
print("Values of 2 unknown variable are as follows:")
print(solve((eq1, eq2), (x, y)))

```

Symbolic Mathematics

Solve the linear equation
 $x + y = 10$
 $x - y = 20$

Dry Run

I/P :- O/P
 Logic

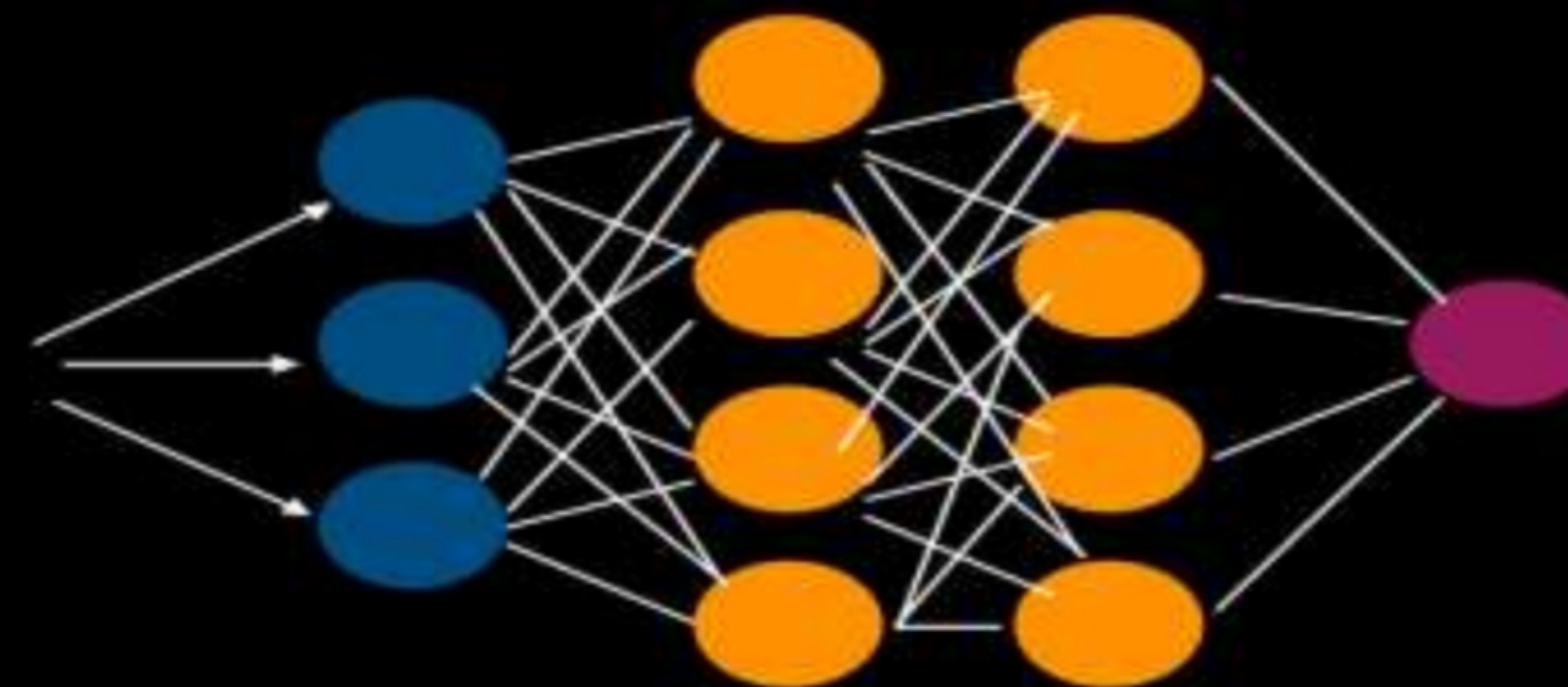
15 -5



ML models Leveraging Linear Algebra



[[12, 213, 44, 15],
[20, 113, 39, 10],
[13, 117, 45, 13],
[25, 120, 49, 14],
...,
...,
],
Tensor



[[0.98, 0.45, 0.44, 0.95],
[0.23, 0.13, 0.39, 0.89],
[0.13, 0.77, 0.55, 0.23],
[0.25, 0.20, 0.49, 0.14],
...,
...,
],
Tensor

Attend the session!

Atelactasis

Ok Google, set alarm for 5 AM.

From Data to Vectors and Matrices



Linear Algebra is **fuel** of machine learning...
Every Type of data can be represented as matrices or as vectors.



→ Summary
→ Practice Problem }

